

HelloWorld_iFlow – External API Integration using SAP CPI

This project demonstrates a basic SAP Cloud Integration (CPI) flow that calls an external REST API and processes the response.

The integration flow represents a client request being processed through SAP CPI, where CPI communicates with an external API, receives a JSON response, and logs the payload using a Groovy script.

Architecture

Client Request → Content Modifier → Request Reply → HTTP Receiver → Groovy Script → End

Testing Note: During implementation in the SAP Trial environment, a Timer trigger was temporarily used to execute and validate the flow due to runtime authorization limitations affecting external HTTP testing.

Step-by-Step Working

1. Client Request

The integration flow begins when a request is sent to CPI.

2. Content Modifier

Prepares or modifies the message before calling the external API.

3. Request Reply

Sends the request and waits for the external API response.

4. HTTP Receiver

Calls the external REST API using HTTP GET.

5. External API Response

Receives JSON payload from JSONPlaceholder API.

6. Groovy Script

Logs the payload into CPI monitoring attachments.

7. Monitoring

Used Monitor → Message Processing to verify successful execution.

External API Used

<https://jsonplaceholder.typicode.com/posts/1>

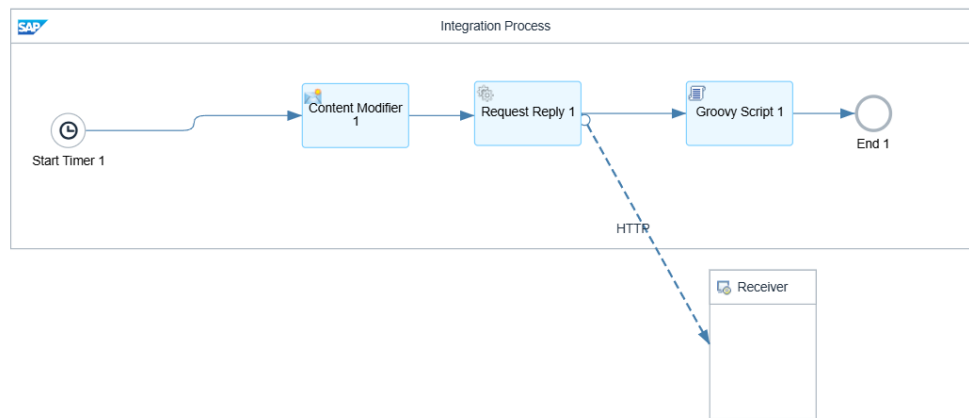
Concepts Learned

- SAP CPI flow structure

- Client-to-API integration flow
- External API communication
- Request Reply processing
- HTTP Receiver adapter
- JSON payload handling
- Monitoring and logging
- Basic Groovy scripting

Screenshots

Integration Flow Design



HTTP Receiver Configuration

HTTP Externalize ?

General **Connection**

CONNECTION DETAILS

Address: *

Query:

Proxy Type:

Method:

Send Body: ☐

Authentication:

Successful Flow Execution

Time: Past Hour

Status: C...

Type: A...

Package: All

Artifacts: All

or

ID: Message, Correlati...

May 11, 2026, 17:06:14 - May 11, 2026, 18...

Messages (2)

<<

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1

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Artifact Name	Status
HelloWorld_iFlow	Completed
May 11, 2026, 17:56:29	1 sec 39 ms
HelloWorld_iFlow	Completed
May 11, 2026, 17:36:54	333 ms

HelloWorld_iFlow

Last Updated at: May 11, 2026, 17:56:29

Status

Properties

Logs

+2

Status

✓

Message processing completed successfully.

Processing Time: 1 sec 39 ms

Properties

Message ID: AGoByvRdGKgNZVqK_NvWVqNCgQon

API Response Payload

HelloWorld_iFlow

Last Updated at: May 11, 2026, 17:56:29

Status: Completed

Log Level: Info

Processing Time: 1 sec 39 ms

Log

Attachments

API Response

Download

```
{
  "userId": 1,
  "id": 1,
  "title": "sunt aut facere repellat provident occaecati excepturi optio reprehenderit",
  "body": "quia et suscipit\nsuscipit recusandae consequuntur expedita et cum\nreprehenderit molestiae ut ut quas totam
}
```

Project Summary

This project demonstrates how SAP CPI communicates with external REST APIs, processes responses, and logs payload data for monitoring and debugging. The same Request-Reply integration pattern is widely used in enterprise integrations involving SAP systems and third-party applications.